Bicol University College of Science
Department of Information Technology

IT106 - Application Development in Emerging Technologies
2nd Semester 2025-2026

Activity:

Group REST API Enhancement and Multi-Client Implementation

Personal Activity Journal

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I. Feasibility and Usability Consultation

I've consulted with Cayla (Front-End Developer) regarding the feasibility and usability of the system's initially planned user interface and workflow. During the consultation, we evaluated whether the proposed design was practical to implement, would fit our time for the development process, and would be user-friendly for the users (teachers and students). We also discussed improvements to enhance navigation, accessibility, and overall user experience.

After the consultation, we conducted a brainstorming meeting to finalize the system's primary pages and core features. We ensured that each feature aligns with the user's needs while remaining feasible within the requirements of this activity.

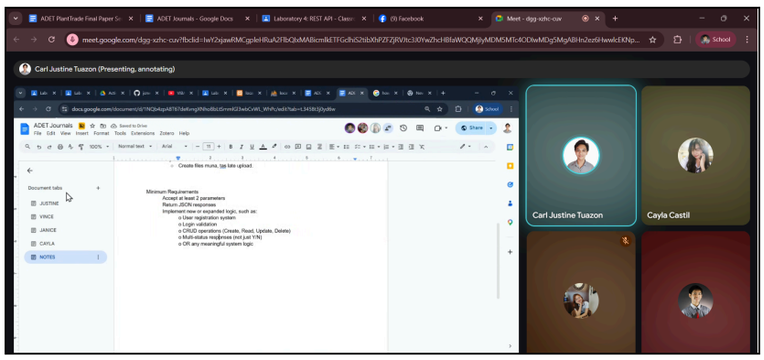


Figure 1. Group Meeting

The identified primary pages or features to be developed include:

- Registration Page
- Login Page
- Teacher UI
    - Course management
        - Create a course
        - View existing courses
        - View students enrolled in a course
- Student UI
    - Course enrollment
    - View courses enrolled in
    - View classmates also enrolled in the course (optional)
    - Unenroll from a course

II. UI/UX Designing Process

I was assigned as the UI/UX Designer. At first, I started with a low-fidelity design of the pages to visualize what the initial layout would look like, based on the identified pages and features we brainstormed during the meeting. These low-fidelity designs focused on basic elements such as the placement of buttons, text fields, navigation menus, and overall page flow without emphasizing visual details.

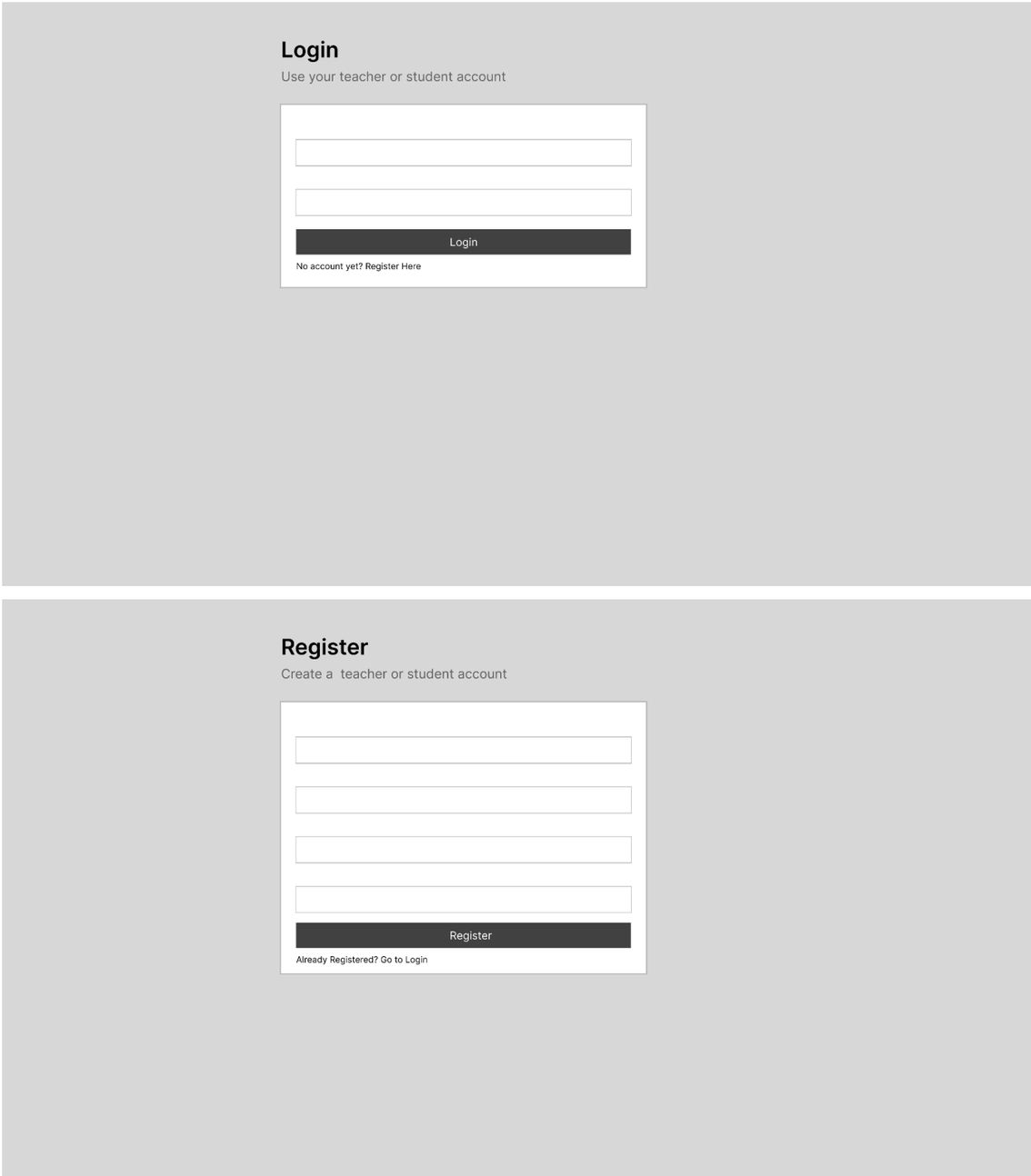


Figure 2.1. Low-Fi Design (Registration / LogIn Page)

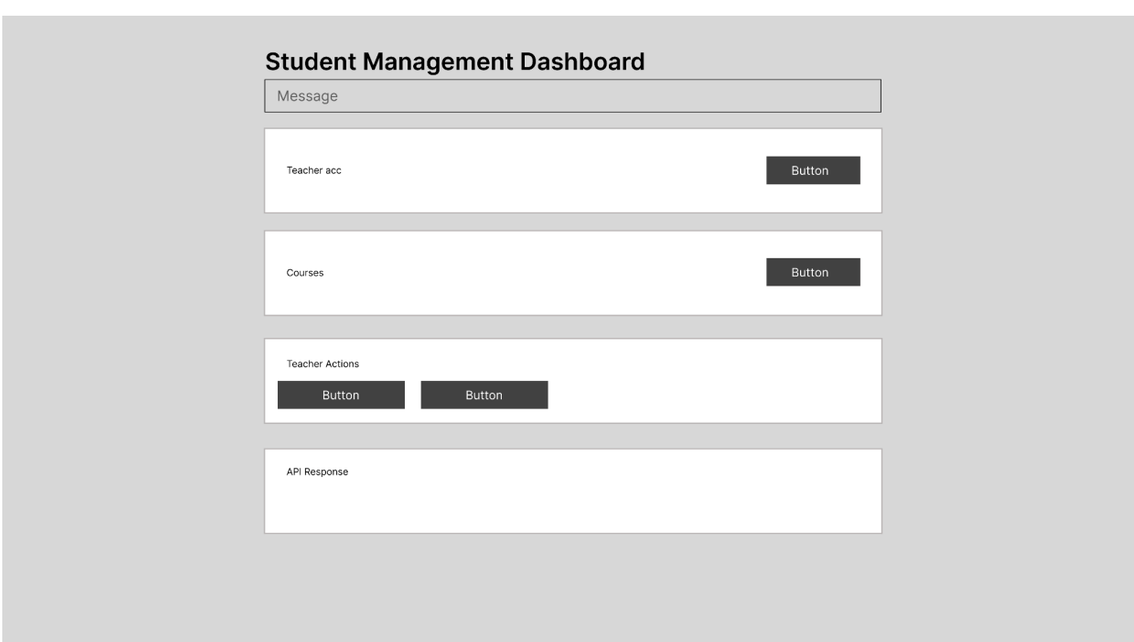


Figure 2.2. Low-Fi Design (Teacher UI)

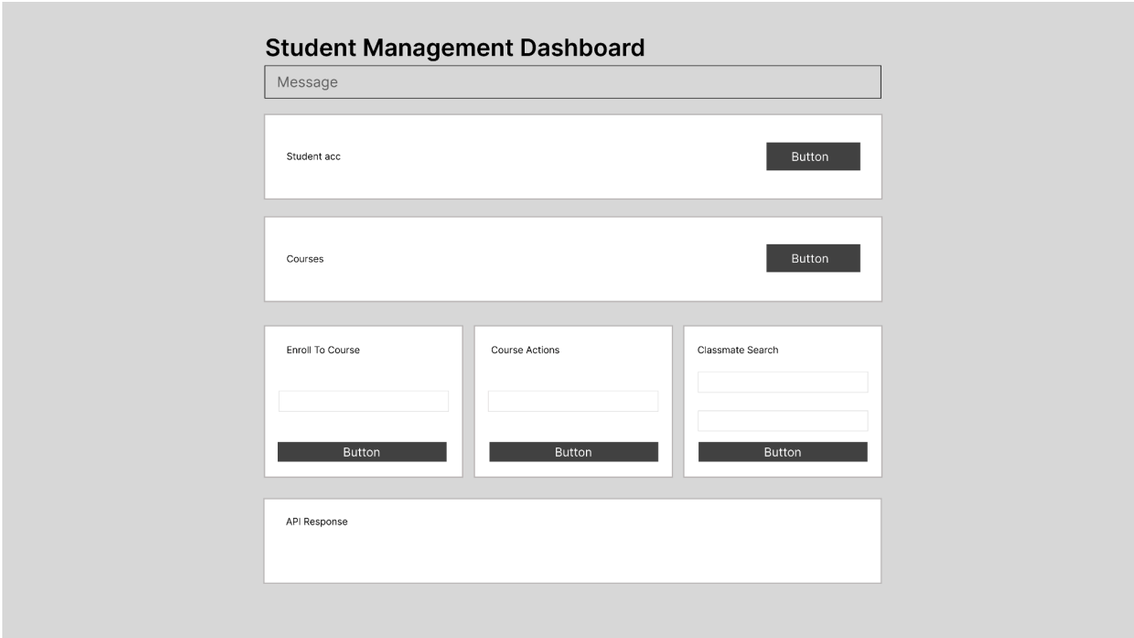
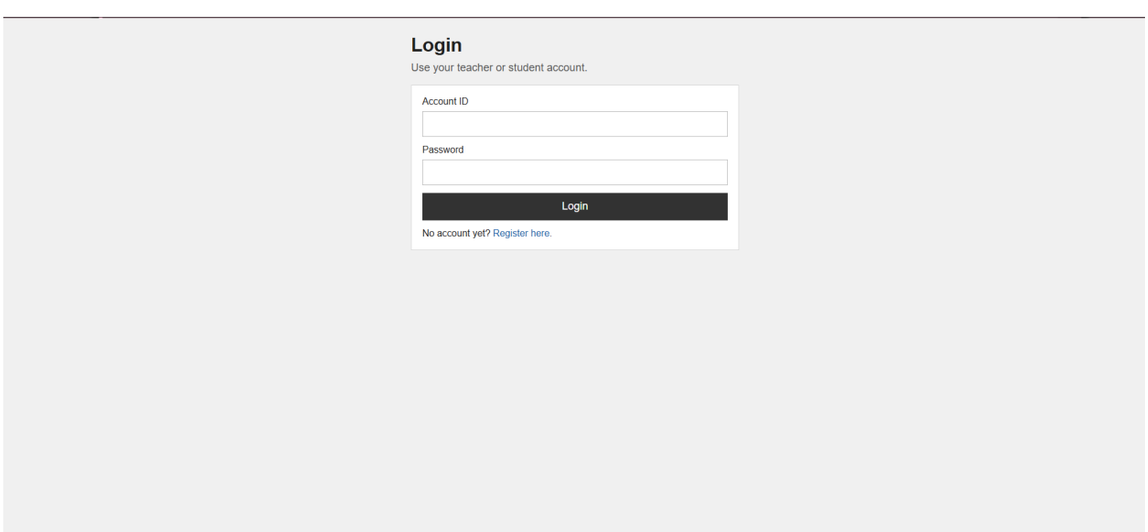


Figure 2.3. Low-Fi Design (Student UI)

After creating the initial designs, I reviewed and refined them to ensure the layout was clear, organized, and easy for users to understand. Adjustments were made to improve navigation and ensure that users can perform tasks efficiently, such as registering, logging in, managing courses, and enrolling in classes.

One of the challenges I encountered was organizing the layout in a way that is both visually appealing and easy to navigate. Another challenge was maintaining consistency across all pages while handling multiple features. To address these challenges, I continuously refined the design and collaborated with the front-end developer, Cayla, to ensure feasibility and usability.

I proceeded to develop high-fidelity designs by incorporating visual elements such as color schemes, typography, spacing, and consistent styling. This stage focused on making the interface more visually appealing while maintaining usability and clarity. I ensured that the design followed a consistent theme across all pages.



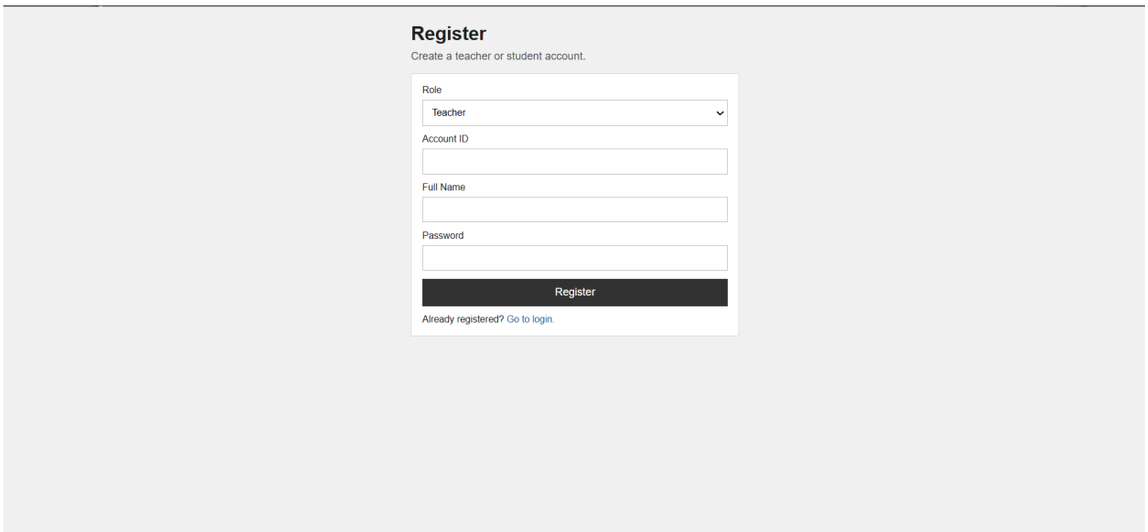


Figure 3.1. High-Fi Designs (Registration / LogIn Page)

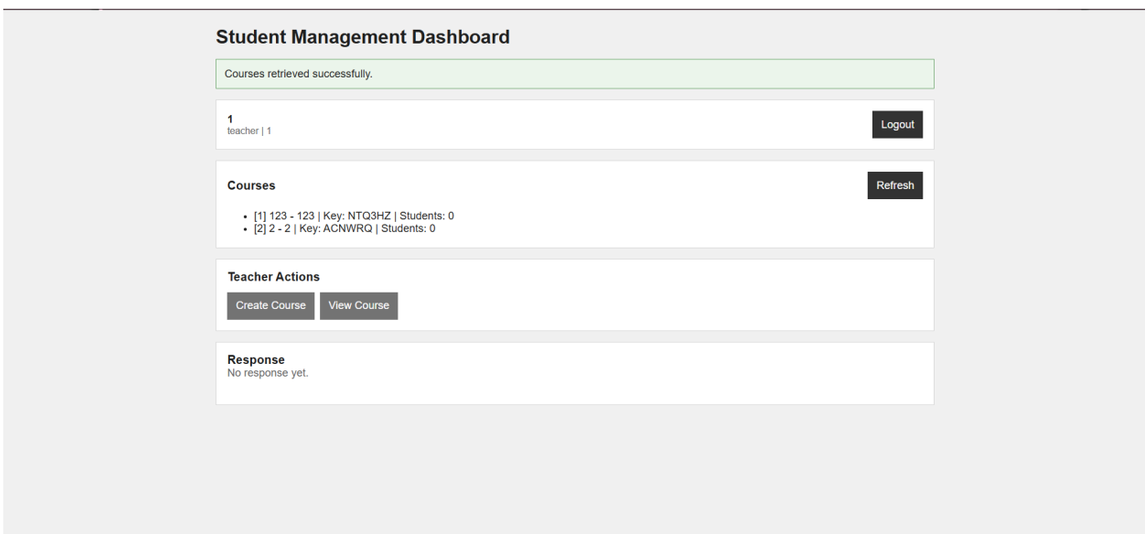


Figure 3.2. High-Fi Designs (Teacher UI)

III. Finalization of UI/UX Designs

After completing the high-fidelity designs, the next step was to finalize the UI/UX designs for implementation. During the finalization process, minor adjustments were made to improve the interface's overall appearance and usability.

Collaboration played an important role in finalizing the designs. I coordinated with Cayla (Front-End Developer) to ensure that the designs were practical to implement. Any design elements that were difficult to develop were adjusted accordingly without compromising usability.

Additionally, I ensured that the finalized designs maintained a user-centered approach by keeping the interface simple, clear, and accessible. All core features were emphasized to make them easy to use and understand, while optional features were included without affecting the main workflow.

The finalized UI/UX designs served as the blueprint for the system's front-end development. This ensured that the system would be implemented according to the planned design and would provide a consistent, user-friendly experience.

IV. Front-End Implementation and Interaction Design

I've collaborated with Justine (Back-End Developer) on specific parts of the source code related to the system's design. This collaboration focused on ensuring that the front-end interface and back-end functionality are properly integrated.

```html
<body data-page="login">
    <main class="page page-narrow">
        <header class="page-header">
            <h1>Login</h1>
            <p>Use your teacher or student account.</p>
        </header>

        <div id="statusBox" class="status hidden" role="status" aria-live="polite"></div>

        <section class="panel">
            <form id="loginForm" class="form-card">
                <label>
                    Account ID
                    <input type="text" name="account_id" required>
                </label>

                <label>
                    Password
                    <input type="password" name="password" required>
                </label>

                <button type="submit">Login</button>
```

Section 1. LogIn Form Code

This figure shows the structure of the login form in the web client. It collects the user's account ID and password via a simple interface and serves as the entry point for both teachers and students.

```javascript
function renderResult(result) {
    // Render the shared line-based response sections returned by the API.
    setStatus(result.message || "Request completed.", result.status === "success" ? "success" : "error");

    if (!elements.responseBox) {
        return;
    }

    const groups = [
        { title: "Courses", lines: result.course_lines },
        { title: "Details", lines: result.detail_lines },
        { title: "Students", lines: result.student_lines },
        { title: "Classmates", lines: result.classmate_lines },
        { title: "Result", lines: result.result_lines }
    ];

    elements.responseBox.innerHTML = "";
    elements.responseBox.classList.remove("empty");
```

Figure 4. Rendering results for the response box

This function is responsible for displaying API responses in the web client. It shows the returned message and organizes additional data, such as courses, details, students, classmates, or result lines, into readable sections inside the response box.